

Article

Fuzzy Neural Network for Detecting Anomalies in Blockchain Transactions

Łukasz Apiecionek ^{1,*}  and Paweł Karbowski ²¹ Faculty of Computer Science, Kazimierz Wielki University, Jana Karola Chodkiewicza 30, 85-064 Bydgoszcz, Poland² Independent Researcher; pauloka09@gmail.com

* Correspondence: lukasz.apiecionek@ukw.edu.pl; Tel.: +48-606-120-335

Abstract: This publication focuses on the use of the artificial intelligence for detecting anomalies, especially in the blockchain network. The research methodology includes the selection of anomalies to be detected and the processing of blockchain data. Various artificial intelligence methods were implemented for anomaly detection as part of the tests, and one new solution—a Fuzzy Neural Network—was presented. The findings indicate the possibility of detecting selected anomalies in the blockchain using artificial intelligence, which is of significant importance for the security of this technology. The conclusions present a discussion on limitations, future research prospects, and guidelines for future work.

Keywords: blockchain; artificial intelligence; anomaly detection; Fuzzy Neural Network

1. Introduction

Blockchain technology is an advanced tool that transforms the functioning of many sectors of the economy by introducing an innovative approach to data storage and transaction execution. It can be described as a distributed database where information is organized into blocks and linked in a chain using special cryptographic techniques.

In traditional systems, such as databases or ledgers, information is stored centrally and managed by a single entity, like a bank or a logistics company. In the case of blockchain technology, data are distributed across multiple computers or nodes that cooperate to form a network. Each node has a copy of the entire transaction history, which means there is no single point of failure and the information is resistant to loss.

An important feature of the blockchain is that each block in the chain contains a reference to the previous block using a unique identifier known as a cryptographic hash. This means that any change in one block will affect the hashes of all subsequent blocks, making unauthorized changes practically impossible. Additionally, blocks are protected by complex mathematical algorithms, which make attacks and unauthorized access difficult.

For transactions, the blockchain allows for the direct exchange of value between parties without the need for a trusted intermediary, such as a bank. Smart contracts, which are programs running on the blockchain, automate and enforce agreements, eliminating the need for third-party involvement. They speed up transaction processes, reduce costs, and increase security since the contracts are coded in an immutable and indisputable manner.

Blockchain technology has many applications in various fields, from finance and logistics to data protection and identity management. It can be used for tracking the provenance of products, ensuring election integrity, sharing patients' medical histories, and even revolutionizing economic systems on a global scale.

Detecting anomalies in blockchain technology is a crucial element in maintaining its integrity, security, and reliability. The blockchain, as a distributed ledger of transactions, relies on the principle of consensus, where network participants must agree on the validity



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and correctness of transactions. However, despite the many benefits this technology offers, there are several potential risks associated with its employment.

One of the primary risks is the possibility of fraudulent transactions. Due to the transparency and permanence of the blockchain, fraudulent transactions can disrupt the system, threatening its integrity. Detecting such anomalies and taking action to stop suspicious transactions is extremely important to ensure the protection of the blockchain system.

Another threat involves attempts to attack the blockchain network. Attackers may try to take control of network nodes, alter transactions, or manipulate data to gain financial benefits or unfairly influence the system's operation. Anomaly detection can help identify irregular activities and take appropriate countermeasures, such as excluding suspicious nodes from the network.

Data manipulation also threatens the blockchain technology. This can include attempts to introduce false information into transaction blocks or modify existing data. Through systematic monitoring of the blockchain network and transaction analysis, such manipulations can be detected, and corrective actions can be taken to restore data integrity.

Data manipulation also threatens blockchain technology. This can include attempts to introduce false information, such as fraudulent transaction records, false identities, or tampered smart contracts, into transaction blocks or modify existing data. Such manipulations could undermine the trustworthiness of the network by allowing unauthorized access or enabling double-spending. For instance, the insertion of false transaction records might lead to disputes in financial or supply chain contexts, eroding confidence in the system's immutability and transparency. Through systematic monitoring of the blockchain network and transaction analysis, such manipulations can be detected, and corrective actions can be taken to restore data integrity.

In order to effectively detect anomalies in blockchain technology, various methods and tools are employed. Statistical analysis, machine learning, and pattern recognition techniques are commonly used to identify abnormal behaviors. Transaction and node monitoring systems can also help in the early detection of suspicious activities.

An important aspect of anomaly detection is also adhering to security principles and protocols. Blockchain systems often utilize mechanisms such as digital signatures, cryptographic algorithms, and consensus protocols, which help verify transaction validity and protect against tampering.

Blockchain technology can be used to secure artificial intelligence mechanisms [1], while it also requires AI mechanisms to detect and prevent anomalies within its network.

In [2], the authors present a general solution that leverages blockchain features to provide an Anomaly Detection Service, enabling peers in a blockchain network to protect against eclipse attacks by sharing information on previous attacks (i.e., by redistributing malicious forks to the entire peer-to-peer community). In [3], the authors introduce a solution named ADvISE, the first Anomaly Detection tool for Blockchain Systems, which leverages blockchain metadata, specifically forks, to collect potentially malicious requests in the network/system while remaining resilient to eclipse attacks. ADvISE collects and analyzes malicious forks to build a threat database, which enables detection and prevention of future attacks.

In [4], a range of algorithms, including isolation forest, histogram-based outlier detection (HBOS), cluster-based local outlier factor (CBLOF), principal component analysis (PCA), K-means, deep autoencoder networks, and ensemble methods are explored for anomaly detection. However, as noted in [5], there is no single optimal approach; it depends on the type and requirements of the specific problem scenario.

The literature also highlights the main methods of attack on blockchain technology, specifically denial of service (DoS), Eclipse, spoofing, and Sybil attacks [6].

Furthermore, real-time anomaly detection is essential, and ready-made solutions for such operations are available in the literature. For example, one framework leverages machine learning algorithms to learn patterns, relationships, and behaviors from historical blockchain data, enabling the real-time detection of anomalous activities [7]. Additionally,

there are proposals to enhance technology by adding more data, such as expanding the concept of the Smart Contract, traditionally assumed to be deterministic, to handle noisy sensor data [8].

Other solutions [9] use mechanisms based on analyzing blockchain network traffic statistics (rather than ledger data) to detect malicious events, through data collection and anomaly detection functions. Nonetheless, deep networks remain one of the most widely used mechanisms [10].

The aim of this publication is to present the results of research on selected methods and techniques for detecting anomalies in blockchain technology and one new proposition, a Fuzzy Neural Network with Ordered Fuzzy Numbers, with a particular focus on analyzing the potential application of artificial intelligence in identifying irregularities in a blockchain network.

2. State of the Art

Blockchain technology has become one of the most innovative and widely discussed tools in the fields of computer science and finance. It has gained popularity due to its unique features, such as decentralization, security, immutability, and data transparency. In this chapter, the definition of blockchain technology is presented, based on various literary sources and experts in this field.

Blockchain technology is a chain of blocks used to store and transmit information about transactions made on the Internet. These transactions are arranged as consecutive blocks of data, with each block containing information about the transaction's value and the time it was made. Each new block is attached to the previous ones, thus forming an unbreakable chain. The condition for adding a block to the chain is finding a "nonce" value (number used once), so that the hash of the entire block meets a specified condition.

A block contains a header with information such as version, timestamp, previous block hash, chain membership, nonce, etc. The element determining the uniqueness of the transaction is the block hash, generated using the proof-of-work algorithm.

Blockchain technology employs cryptographic solutions, such as one-way hash functions and asymmetric cryptography. This ensures the system's security and eliminates the issue of trust, without the need for a central supervising institution. This approach makes blockchain technology groundbreaking and resistant to manipulations, making it attractive to companies and institutions that prioritize security, confidentiality, and data integrity [11].

Blockchain has been described as a "digital ledger" [12] or as a "distributed, immutable ledger of data protected by cryptography" [13].

Anomalies in the context of blockchain technology refer to irregularities or inconsistencies that occur within the blockchain system, potentially affecting the integrity, security, and immutability of the data stored in the network.

In order to minimize the risk of anomalies, many blockchains implement various security mechanisms and protocols. For example, Proof-of-Work (PoW) and Proof-of-Stake (PoS) systems are designed to make 51% attacks and Sybil attacks more difficult. Additionally, cryptographic techniques, such as digital signatures and hashing, are used to ensure data integrity.

Anomalies in blockchain technology are incorrect or unexpected events that can occur in the blockchain network [14]. One of the most commonly encountered anomalies is double-spending, which happens when an entity in the network spends the same funds twice, leading to a loss of transaction integrity. Another type of anomaly is a 51% attack, which involves gaining control of 51% of the computational power in the blockchain network and allows the attacker to manipulate transaction history or prevent other participants from confirming transactions.

A scholarly article presents a comprehensive analysis of various anomalies in blockchain technology [15]. The authors propose a taxonomy of anomalies, encompassing both techni-

cal and social anomalies. They also analyze the causes of these anomalies, including design errors, implementation errors, user behaviors, and external attacks.

Key anomalies in blockchain technology are the following:

- Double-Spending: This occurs when an individual spends the same digital currency unit more than once, compromising the integrity of transactions.
- 51% Attack: This type of attack happens when a single entity or group controls more than 51% of the network's power hash rate or staking power, enabling them to alter the blockchain's transaction history or halt new transactions from being confirmed.
- Technical Anomalies: These can be caused by bugs or flaws in the blockchain's design or implementation, potentially leading to vulnerabilities that attackers can exploit.
- Social Anomalies: These are related to behaviors and interactions of users within the blockchain network, including trust issues and fraudulent activities.
- External Attacks: These involve external parties' attempts to disrupt the blockchain network through various means, such as distributed denial-of-service (DDoS) attacks or exploiting vulnerabilities in network protocols.

It is possible to detect such anomalies and take corrective actions to restore data integrity by systematically monitoring the blockchain network and analyzing transactions. This often involves the use of statistical analysis, machine learning, and pattern recognition techniques to identify abnormal behaviors. Transaction and node monitoring systems also play a crucial role in the early detection of suspicious activities [16]

Overall, understanding and detecting anomalies in blockchain technology are essential for maintaining the security, reliability, and integrity of blockchain systems, ensuring all that was above described allows the systems to remain robust and trustworthy for all users [17].

2.1. Existing Methods for Detecting Anomalies in Blockchain Technology

Books, scholarly articles, and practical studies dedicated to blockchain technology often discuss these challenges and propose various data analysis methods that can help detect anomalies and suspicious transaction patterns.

One of the key sources of information regarding anomaly detection can be pattern analysis. The potential violations can be identified by utilizing statistical analysis techniques; transactions that significantly deviate from the norm or exhibit suspicious behavior patterns may indicate potential violations. There are also techniques using machine learning that can automatically detect anomalies based on historical patterns. For example, classification and clustering algorithms can be used to identify irregularities such as suspicious transactions or activities of suspicious users [18–20].

Smart contracts can be programmed to analyze data and transactions in real-time, identifying potential anomalies and suspicious behavior patterns [21]. The authors of the publication propose using classification and clustering algorithms to identify suspicious transactions and user behaviors [22].

Other methods of detecting anomalies in blockchain technology include social network analysis, where connections between different wallet addresses are examined to identify suspicious patterns of activity. There is also an approach based on utilizing consensus mechanisms, where the blockchain network is monitored for irregularities in the block creation process.

It is also important to note that the development of blockchain technology and anomaly detection methods is in constant progress. New research and innovations emerge regularly, and the literature on this topic is constantly being updated.

2.2. Gaps and Challenges in Existing Anomaly Detection Methods

Blockchain technology, while offering many benefits, unfortunately also suffers from numerous difficulties and issues, particularly in the realm of anomaly detection:

- Lack of standardization—Uniform standardization in the data analysis process in blockchain technology poses one of the key challenges in implementing solutions [23].

- **Complexity of Transaction Analysis**—Analyzing transactions on the blockchain is a complex process due to the vast amounts of data generated by each blockchain network and the intricate patterns of interaction between different wallet addresses. Existing analysis methods often rely on visualization or heuristic techniques, which, unfortunately, can lead to certain limitations and errors in the anomaly detection process [24].
- **Lack of Data Labeling**—The absence of access to labeled data poses a significant challenge, especially in machine learning, where models require training data that are clearly labeled as anomalies and normal data. In the realm of blockchain technology, such labeled data are often unavailable or difficult to access, introducing an additional level of complexity in the process of training effective models [25].
- **Dynamicity of Blockchain Networks**—Blockchain networks are incredibly dynamic and continuously evolving. New blocks are added to the chain, and transactions are executed in real time. This dynamic nature poses a significant challenge for anomaly detection methods, which must be able to adapt to the continuous changes in the network's structure and operation [26].
- **Technological Development and Attacks**: The blockchain technology is dynamic and constantly evolving, and the same applies to potential threats from attackers. This necessitates ongoing adaptation and development of anomaly detection methods in blockchain networks to meet new challenges and the increasing sophistication of attacks [27].
- **Limited Computational Resources**—One of the key aspects raised in research is that the size of blockchains continually grows with the number of transactions and new blocks added to the network [28]. This increasing volume of data presents a challenge for blockchain analysis, as it requires increasingly complex computations. Standard personal computers may become inadequate for conducting sophisticated analyses, leading to prolonged and costly computational processes.
- **Privacy and Anonymity**—The blockchain, as a technology based on a decentralized and transparent accounting system, introduces a certain level of privacy and anonymity for network participants. Each participant is identified by a wallet address rather than a name or personal data. This means that transactions are conducted without revealing the identities of the parties involved in the operation. However, the same features that make blockchain attractive to users can pose a challenge for anomaly detection, especially in case where attackers attempt to operate anonymously [29].
- **Limited Social Acceptance**—Anomaly detection methods on the blockchain have features that raise concerns regarding user privacy and undermine the social acceptance of this technology. With the development of blockchain analysis, it becomes clear that collecting and analyzing data to detect anomalies may violate the privacy of participants. Efforts in this area emphasize the need to find a balance between ensuring network security and respecting user privacy [30]. Privacy is particularly significant in the financial services sector, where blockchain is gaining popularity. Sharing financial data for analysis purposes may raise concerns among users and regulatory oversight authority. However, concurrently, it is essential to maintain the security and integrity of the blockchain network, especially in the context of preventing financial fraud and money laundering. Deliberations on the balance between security and privacy in blockchain analysis are crucial for the long-term success of this technology. As the blockchain evolves and finds increasingly wider applications, there is a need to develop solutions that allow for effective anomaly detection without compromising participant privacy. This is a challenge that requires further research and the development of more advanced blockchain analysis tools.

3. Methodology

3.1. Selection of Anomalies for Detection in Blockchain

For the purposes of this research, five key anomalies were selected, which appear to be promising and valuable in the context of anomaly detection: (1) Sybil attacks at the address level, (2) double-spending, (3) anomalies in smart contracts, (4) analysis of transactions with a large number of confirmations equal to zero, and (5) high-frequency transaction anomalies.

Sybil attacks at the address level represent a sophisticated technique aimed at subtly introducing misinformation and disrupting the participation structure in the system, especially in the context of a blockchain [31]. In this attack, the aggressor creates fake identities or addresses, simulating the existence of multiple independent entities. In the case of a blockchain, the Sybil attack at the address level manifests itself through the use of a strategy in which one entity pretends to be a series of different users, while simultaneously controlling all fake identities. As a result of this action, misinformation is introduced into the blockchain network, disrupting the true picture of the participation structure. For example, an attacker can create a large number of fake addresses, thereby suggesting a much larger presence in the network than is currently true. Such manipulation can mislead other participating entities and make it difficult to detect any anomalies or irregularities. In the context of research on Sybil attacks at the address level, it is important to focus on developing effective methods for identifying fake addresses [32].

Blockchain transaction analysis may involve various methods, such as in-depth analysis of transactional patterns, anomaly detection in the transaction graph structure, or monitoring activities of suspicious addresses. It is worth noting a characteristic feature of Sybil attacks in the context of blockchain technology. Specifically, one of the clear signs of these types of attacks at the address level in a blockchain is the unusual situation where the transaction sender's address is exactly the same as the recipient's address. This unusual and atypical behavior is a key element that can be used as a significant reference point when creating effective countermeasures and defensive strategies against such attacks. In practice, this means that during a Sybil attack, the address sending the cryptocurrency is identical to the address receiving it. Such a situation is extremely unusual because, under normal circumstances, different addresses correspond to different transaction participants. Detecting this anomaly allows for quick identification and response to potential threats. Developing effective countermeasures and defensive strategies against this type of attack is crucial for maintaining the security of blockchain-based systems.

Double-spending represents a significant challenge in the context of cryptocurrencies, especially when using blockchain technology. This problem involves the potential ability to use the same units of cryptocurrency multiple times. Therefore, ensuring transaction security becomes a crucial aspect of the functioning of blockchain-based systems [33]. Research on double-spending requires a detailed analysis of various scenarios related to this threat. It is important to identify different techniques and strategies that potential attackers may use to deceive the system and perform a double transaction. For example, one can examine situations where attempts are made to initiate two transactions, spending the same units of cryptocurrency, while deceiving the system [34]. In the context of research on double-spending, it is also important to consider the possibilities of using artificial intelligence in detecting attempts of double-spending. Analysis of transaction patterns, the use of machine learning techniques, and anomaly detection algorithms may prove to be crucial for effectively eliminating this type of threat [19].

Anomalies in smart contracts, which are programmable pieces of code in blockchain technology, play a crucial role in ensuring autonomy and executing functions in a decentralized manner. However, with the growing popularity of these contracts, researchers are paying attention to potential anomalies in their operation, which can lead to serious financial consequences and affect the integrity of the entire blockchain ecosystem. Research on anomalies in smart contracts focuses on detecting situations where these contracts exhibit unexpected results or are abused. In the context of research, a detailed analysis

of the source code of smart contracts becomes an indispensable tool. It allows for the identification of potential programming errors, security vulnerabilities, or implementation inconsistencies [15].

With respect to analysis of transactions with a large number of near-zero confirmations, research on transactions with minimal confirmations in the context of securing the blockchain network represents an area of particular importance, where detailed analyses aim to enhance the integrity and security of the system. When transactions receive a significant number of confirmations in a short time, it becomes necessary to gain a deeper understanding to identify potential threats to the system. These studies involve a microscopic analysis of individual transactions, while also developing a macroscopic understanding of the context, history, and dynamic changes over time. Thorough verification of transaction participants, identification of senders and recipients, and assessment of their reputation and previous activities within the blockchain network become key research elements.

Regarding anomalies related to high-frequency transactions, these anomalies, manifesting through excessive transaction activity by a specific address within a short time frame, have become a subject of increasingly in-depth analysis in the context of the complexity of the cryptocurrency market. This unique phenomenon, characterized by extremely intense activity, not only attracts attention due to its potential impact on the market but also opens the door to a full understanding of the dynamic interactions between blockchain participants. In advanced research on the anomaly of high-frequency transactions, a thorough analysis of the interrelationships between addresses plays a crucial role. This detailed context allows for a more comprehensive understanding of why a given address generates an excessive number of transactions. Equally important is focusing on identifying the potential effects of such intense activity, including possible market manipulations or suspicious financial practices.

3.2. Preparation and Processing of Blockchain Data

In order to conduct analysis and detect anomalies in blockchain technology, it was necessary to prepare the appropriate data. Unfortunately, publicly available blockchain data files with labeled anomalies were not accessible, so it was necessary to generate our own dataset using the Python programming language. The goal of data generation was to create a diverse set of blockchain transactions, encompassing various scenarios. The data were generated using a Python script that leverages the “random” library to generate random values, “datetime” to create dates, and “csv” to save the data to a CSV (Comma-Separated Values) file. The main stages of the data generation process are described below:

- **Ethereum Addresses:** In the data generation process, a Python script was used to create random Ethereum addresses, representing both senders and receivers of transactions. By using appropriate random functions, the script generated unique addresses assigned to either the sender or receiver role within blockchain transactions.
- **Date and Time:** The script was configured to set a random date and time for each transaction, limited to the past 3 years. This ensures that the generated transactions have various dates and times within this period, reflecting different moments in time over the last three years (2020–2023).
- **Transaction Amount:** A random amount was assigned to each transaction, ranging from 0.01 to 1000.0 units. Using appropriate random functions, the script generates diverse amount values for individual transactions, resulting in a varied dataset of transaction amounts.
- **Transaction Number:** A unique transaction number was assigned, simulating a transaction ID (identifier) on the blockchain. Using appropriate mechanisms to ensure uniqueness, the script generates transaction numbers that serve as simulated transaction IDs. Each transaction receives its number, allowing for tracking and identification within the dataset.

- **Transaction Status:** A random transaction status was assigned, taking one of three following possible labels: PENDING, COMPLETED, or FAILED. Using appropriate random functions, the script assigns various statuses to transactions, simulating different outcomes of these operations. This process generates data reflecting different scenarios of transaction completion within the analyzed dataset.
- **Transaction Type:** A random transaction type was assigned (SEND, RECEIVE, TRANSFER). Using appropriate random functions, the script assigns various types of transactions, simulating different kinds of operations within the dataset. This process generates data reflecting different types of transactions in the analyzed blockchain dataset.
- **Transaction Fee:** A random transaction fee was assigned, ranging from 0.001 to 0.1 units. Using appropriate random functions, the script generates varied fee values assigned to individual transactions, resulting in a diverse dataset of transaction fees. This process simulates different fee scenarios within the blockchain data.
- **Block Number:** A random block number was assigned to each transaction. Using appropriate random functions, the script generates unique block numbers assigned to individual transactions, resulting in a diverse dataset of block numbers. This process simulates different positions of transactions within the blockchain, allowing for the analysis and testing of anomaly detection algorithms in various block contexts.
- **Confirmations:** A random number of confirmations was assigned to each transaction. Using appropriate random functions, the script generates diverse confirmation values assigned to individual transactions. This random number of confirmations simulates the level of certainty achieved by a transaction within the blockchain network, allowing for the analysis of various confirmation scenarios within the dataset.
- **Smart Contract Address:** A random smart contract address was assigned to transactions. Using appropriate random functions, the script generates unique smart contract addresses assigned to individual transactions. This process simulates different scenarios of using smart contract addresses within the analyzed blockchain dataset, enabling the testing and evaluation of anomaly detection algorithms in various smart contract contexts.
- **Data:** We generated random data representing additional transaction information. These included various attributes or details not directly related to the previously mentioned aspects of transactions, such as comments, session IDs, or other metadata. Using appropriate random functions, the script generates varied additional data assigned to individual transactions, enriching the dataset and allowing for more comprehensive analysis.

By following these steps, a diverse and comprehensive dataset was generated that could be used for detailed analysis and testing of blockchain anomaly detection methods. During the method learning and testing, there were over 500 records and they were divided into 80/20%. As mentioned, the prepared code generates realistic blockchain transaction data by creating unique Ethereum addresses in the “0x” format with hexadecimal numbers, resembling actual wallets. Transaction dates were randomly selected from the past three years to mimic transaction history. Transaction amounts range from 0.01 to 1000, reflecting typical values on the blockchain network. The transaction status was set to “PENDING”, “COMPLETED”, or “FAILED”, just like in real cases.

To enhance the credibility of the data, the code also generated 16-character transaction IDs resembling real IDs, random transaction fees typical of the blockchain, and a confirmation count from 0 to 100, reflecting real network variability. These details make the transactions in the data features appear realistic.

In order to simulate various transaction scenarios, modifications to the generated data depending on the type of anomaly were introduced. In the case of anomalies related to smart contracts, a “00” suffix was added to the generated ETH address, which resulted in a 42-character address instead of the standard 40, which was distinguished and indicated a potential irregularity. For high-frequency transaction simulations, records were generated at short time intervals to simulate the rapid addition of multiple transactions in a brief

period. This approach allowed for the replication of specific situations and typical behavior patterns found in real transactions.

Smart contract addresses were used to simulate various transactional contexts, as they allowed for the creation of specific anomalies, such as unusual address lengths or unique character patterns (such as those ending in “00”). For instance, by modifying the endings of ETH addresses, transactions were generated that stood out among others. However, the block number was not a crucial element in the anomaly detection process, so it did not have a direct impact on detecting unusual transactions.

3.3. Implementation of Existing Anomaly Detection Algorithms

In the research conducted, the following five artificial intelligence algorithms were implemented, specifically tailored to detect anomalies in blockchain transactions:

- **RandomForestClassifier:** The RandomForestClassifier is an advanced machine learning algorithm based on the concept of random forests, which consist of many decision trees. In this context, each tree represents an independent model, and the aggregate of their diverse decisions is combined to create a more robust mechanism for detecting anomalies in blockchain transaction data.
- **GradientBoostingClassifier:** The GradientBoostingClassifier is an advanced machine learning algorithm that utilizes decision trees sequentially to improve model accuracy. Unlike RandomForest, where trees operate independently, GradientBoostingClassifier focuses on progressively correcting the errors of previous trees, leading to the construction of a more precise predictive model.
- **k-means Algorithm:** The k-means algorithm is an advanced data clustering technique aimed at dividing a dataset into k clusters, where each cluster contains elements with similar characteristics. In the context of blockchain transaction analysis, k-means plays a crucial role by identifying groups of transactions with similar properties, which in turn facilitates the detection of unusual patterns among these groups.
- **One-Class SVM (Support Vector Machine):** The One-Class SVM is an advanced machine learning algorithm that excels at training solely on data considered normal. In the context of anomaly detection in blockchain transactions, this model can be adapted to identify cases among data deemed standard.
- **Autoencoder:** An autoencoder is a specific type of neural network used for dimensionality reduction while simultaneously representing the key features of the data. In the context of detecting anomalies in blockchain transactions, the autoencoder plays a pivotal role by enabling the representation of normal transaction (without any anomalies) patterns and facilitating the identification of outlier observations.

Using the k-means algorithm, transactions were grouped based on their features, and clusters were subsequently analyzed for unusual patterns. For example, if a smart contract address contains 42 characters instead of the standard 40, it could be assigned to a cluster with other similar anomalies.

A One-Class SVM model was created with the parameter ν set to 0.1, which controls the proportion of data to be treated as anomalies. Adjusting this parameter is crucial and may require further calibration depending on the data characteristics.

An autoencoder was used to detect anomalies in transactions with fewer than one confirmation. The focus was first placed on these transactions, and the data were standardized, improving model efficiency. The autoencoder was trained for 100 epochs and then used to predict outcomes and calculate mean squared errors (MSEs). By setting a threshold based on the MSE, anomalies were identified.

3.4. Fuzzy Neural Network with Ordered Fuzzy Numbers

Artificial Neural Networks, including Deep Networks, have become highly popular tools in the field of artificial intelligence. These networks consist of interconnected neurons, with the McCulloch–Pitts neuron being a well-known example. Each neuron can receive multiple inputs, each with an associated weight, and generates a single output. Professor

Witold Kosiński and his team expanded on the concept of fuzzy numbers by introducing trends, which have proven valuable for analyzing various events. An Ordered Fuzzy Number is represented as an ordered pair $A = (f_A, g_A)$ where both are continuous functions $F : f_A, g_A : [0, 1] \rightarrow R$ referred to as follows:

- up_A —the initial, rising slope;
- $down_A$ —the final, falling slope.

Although fuzzy networks have been examined in previous studies [32], none have applied Ordered Fuzzy Number (OFN) arithmetic in Artificial Neural Networks. OFNs were selected for neural network use because of the following:

- They allow for computations while overcoming the limitations of traditional L-R numbers;
- No prior efforts have been made to create a network using this approach, making it innovative.

For the construction of a fuzzy network, a modified McCulloch–Pitts neuron model was suggested. However, rather than using standard numerical values, the inputs, weights, and outputs are expressed using OFN notation [33,34]. This notation helps to deal with uncertainty and imprecision in numerical data. Arithmetic operations with OFNs follow two main steps:

Step 1: Multiply each input value by its corresponding weight, then add the products to obtain the final result:

$$S = W_0 + \sum_{i=0}^n X_i W_i, \quad (1)$$

Step 2: Then, the output using OFN arithmetic is calculated:

$$Y = f(S), \quad (2)$$

The S , W , X , and Y elements are represented as values in OFN notation. This approach requires several key layers to be implemented:

- First layer: A fuzzification process transforms the input data into OFN notation.
- Last layer: Defuzzification is applied to process the output data from the network.
- Deep layer: Learning and training algorithms are adapted to function efficiently with OFN arithmetic in the network's layers.

Within the proposed architecture, a deep fuzzy network is used, where traditional neurons are replaced with fuzzy neurons. These neurons have weights represented by directed fuzzy numbers, and the arithmetic of these numbers is utilized for calculations. The first layer handles the fuzzification of the input data, which is a critical step for the practical use of this solution. The fuzzification process and data representation in the fuzzy domain must be tailored to each specific implementation. In the case of multidimensional data, appropriate fuzzification methods will need to be developed. However, many existing solutions can be leveraged to develop data fuzzification methods for further analysis, so this process should not pose significant challenges. The deep layers are responsible for generalizing the information. For this research, a deep fully connected neural network (FFN) has been implemented. The final layer carries out the defuzzification process, and numerous methods for defuzzifying results are available in the literature, facilitating the conversion from fuzzy numbers back to real numbers.

This network was used as a sixth solution for blockchain anomaly detection.

4. Results

In this chapter, the results of the conducted research will be discussed in detail, with the primary aim of examining the effectiveness of the applied artificial intelligence methods in detecting anomalies in blockchain transactions. Each of the selected anomalies was thoroughly analyzed using a dedicated artificial intelligence method implemented in the Python programming language. The analysis of the experimental results is presented in Table 1 and presents the effectiveness of the applied artificial intelligence methods

in detecting various types of anomalies in blockchain transactions. The accuracy was measured as per the following formula:

$$Accuracy = \frac{TruePositive + TrueNegative}{TruePositive + FalsePositive + TrueNegative + FalseNegative} \quad (3)$$

Table 1. Results of individual anomaly detection algorithms.

Anomaly Type	Sybil Attack	Double-Spending	Smart Contracts	Analysis of Transactions with a Large Number of Confirmations Equal to Zero	High-Frequency Transaction Anomaly
Method name	Gradient Boosting Classifier	Random Forest Classifier	k-means	Autoencoder	OneClass SVM
Time (s)	39.69	85.16	0.52	21.40	28.00
Accuracy (%)	94.6	92.1	96.6	97.2	93.9
Anomaly detected [%]	30.27	29.05	29.89	30.32	29.02

The experimental analysis of anomaly detection in blockchain transactions reveals plenty of opportunities and challenges associated with the diversity of anomalies. In the temporal context of experiments, the k-means method stands out in terms of its efficiency in time-sensitive operations, emphasizing its potential application in situations requiring an immediate response to anomalies. However, each analyzed anomaly requires an individual approach, introducing complexity to the detection process and posing challenges for researchers due to the diversity of anomaly types.

The effectiveness of the Gradient Boosting Classifier in identifying Sybil attacks may be due to its ability to dynamically adapt to variable conditions. In the case of double-spending, the Random Forest Classifier method demonstrates solid accuracy, suggesting that this model effectively identifies such fraudulent activities. Nevertheless, each method has its strengths and limitations, underscoring the need for further contextual analysis.

The processing time, as a key parameter of practical applications, confirms the efficiency of the k-means method, especially in situations requiring an immediate response to anomalies. The analysis of model errors, considering false positive and false negative rates, provides valuable insights into areas where each method could be further optimized.

Expanding the perspective to the results of using Autoencoder and OneClassSVM, we can note that Autoencoder, as an unsupervised learning technique, proved effective in identifying anomalies, as evidenced by the detection rate of 29.02%. On the other hand, OneClassSVM, as a one-class learning method, achieved a 30.32% anomaly identification rate, further highlighting the richness of possibilities in detecting anomalies in blockchain transactions.

Analyzing these additional results, we can see that each method has unique characteristics that are valuable to consider in the context of further research. A comprehensive understanding of the results requires deeper contextual analysis and consideration of the diversity of anomaly types occurring in this specific domain. Meanwhile, exploring different detection methods opens up new perspectives for the developing blockchain technology.

The results achieved by the Fuzzy Neural Network with Ordered Fuzzy Numbers are presented in Table 2.

The results of the analysis in the area of anomaly detection in blockchain transactions provide a solid foundation for continued research and refinement of existing methods. In light of the accumulated knowledge, there is an opportunity to tailor detection models to the specialized requirements of various anomaly scenarios.

Table 2. Results of Fuzzy Neural Network.

Anomaly Type	Sybil Attack	Double-Spending	Smart Contracts	Analysis of Transactions with a Large Number of Confirmations Equal to Zero	High-Frequency Transaction Anomaly
Method name	Fuzzy Neural Network	Fuzzy Neural Network	Fuzzy Neural Network	Fuzzy Neural Network	Fuzzy Neural Network
Time (s)	37.5614	64.2014	45.6816	16.0034	18.1923
Accuracy (%)	93.8	91.1	95.1	97.72	91.0
Anomaly detected [%]	34.27	29.05	32.14	30.32	27.02

The proposed Fuzzy Neural Network with Ordered Fuzzy Numbers achieved results comparable to existing methods. It is worth noting that the employed solution was faster in all experimental trials.

5. Discussion

In the conducted research, various artificial intelligence methods were applied to identify different types of anomalies in blockchain transactions. Threats such as Sybil attacks, double-spending, smart contracts, transactions with a large number of confirmations close to zero, and high transaction frequencies were analyzed. The results obtained in the experiment confirm that artificial intelligence techniques are effective in detecting irregularities in the discussed scenarios.

In terms of the effectiveness of individual methods, Gradient Boosting Classifier and Random Forest Classifier achieved accuracy rates of 94.6% and 92.1%, respectively. Despite its short execution time (0.52 s), the k-means method impressively achieved an accuracy rate 96.6%. The Autoencoder achieved an accuracy rate of 97.7%, confirming its effectiveness in identifying anomalies in blockchain transactions.

The proposed Fuzzy Neural Network with Ordered Fuzzy Numbers demonstrated similar accuracy and effectiveness in anomaly detection while performing the task faster. Therefore, it is a very promising solution for anomaly detection in blockchain networks, where high speed is crucial. The Random Forest Classifier has several drawbacks: high computational complexity, lack of sensitivity to rare anomalies, tendency to overfit, and difficulties with interpreting results. The Gradient Boosting Classifier also comes with high computational demands, sensitivity to noise in the data, the need for extensive hyperparameter tuning, and a tendency to overfit. The k-means algorithm has its own issues, including assumptions of spherical clusters, sensitivity to the choice of the number of clusters, difficulties in detecting anomalies, and susceptibility to noise and outliers. The One-Class SVM method suffers from high computational complexity, dependence on the optimal choice of kernel, scalability problems, and sensitivity to hyperparameters. Finally, Autoencoders are challenged by model complexity, the need for large amounts of data, the risk of overfitting, lack of interpretability, and sensitivity to hyperparameter tuning.

The proposed method requires preparing labeled data to carry out the training process. To apply the method, data must first be collected and then labeled, divided into training and testing sets, and then the training process conducted. Therefore, the method is not sensitive to the type of blockchain network chosen.

The advantage of using the Fuzzy Neural Network with Ordered Fuzzy Numbers is that it is one technology that gives almost the same results as the others. So, it is easier for an expert to prepare it, learn it, and provide the results. The implementation process should also be quicker because it requires using only one technology—i.e., one library. This is important as a practical matter when it comes to real implementations.

The Gradient Boosting Classifier method generally has higher computational complexity than Random Forest, as it is an iterative algorithm where each subsequent tree is built to improve the results of the previous ones. Although it offers very high accuracy in anomaly detection, it has a relatively high prediction cost. Using this solution in real-time may require optimization. On the other hand, the Random Forest Classifier has lower

computational complexity than the Gradient Boosting Classifier, since each tree is built independently. It can also make predictions faster, as trees can be searched in parallel. Thanks to parallel processing, Random Forest can be used for anomaly detection with an acceptable response time. It performs well with smaller datasets and cases where rapid response is necessary. However, for very large input data, it may be necessary to limit the number of trees or reduce their depth.

The k-means method is relatively simple computationally compared to the other methods. It is highly efficient both in the training and prediction phases, making it suitable for real-time applications, especially for detecting outlier clusters. However, this method may be less accurate with complex, high-dimensional data typical of blockchain networks.

Autoencoder training has a computational complexity similar to neural networks and requires significant computational resources, especially with large networks. Autoencoders are well-suited for detecting nonlinear dependencies, which is useful in complex networks such as blockchains. Autoencoders can operate in real-time.

One-Class SVM has quite high computational complexity, making it inefficient for large datasets. It also has high memory requirements, which can be a limitation. It performs well with small-to-medium-sized datasets but may be computationally prohibitive for very large datasets, excluding its use in real-time scenarios. In the conducted experiments, One-Class SVM was used with only timestamps as features, which means the dataset contained a single measurement variable (number of variables: one). The dataset consisted of 50,000 samples. The prepared data were processed into an appropriate time-based format to ensure compatibility with the model requirements. The model was configured so that approximately 30% of the samples could be classified as anomalies ($\nu = 0.3$). The results were recorded as 1 for normal operations and -1 for anomalies.

The complexity of a Deep Neural Network (DNN) depends on the number of neurons, and the training process is influenced by the number of epochs, data, and neurons. Deep networks are highly resource-intensive, requiring significant computational power, especially with complex architectures. GPUs are often used to accelerate calculations.

The proposed Fuzzy Neural Network, in preliminary experiments, demonstrated lower architectural complexity and a faster learning process than deep networks, but overall conclusions in this area require further research.

Tree-based models, such as Random Forest, are generally more interpretable than neural networks, as they provide direct insights into the decision-making process and allow for easy explanation of feature impact on the outcome. However, the mechanisms used in the proposed Fuzzy Neural Network require further work to improve interpretability. The authors recognize the need for ongoing research in this area.

6. Conclusions

The achievements presented in this publication have focused on confirming the potential of artificial intelligence techniques in detecting anomalies in blockchain networks, especially using the proposed Fuzzy Neural Network with Ordered Fuzzy Numbers. The experimental results provide significant evidence of the effectiveness of the applied methods in identifying various deviations from the norm, which perfectly aligns with the main goal of this work.

Furthermore, the accomplished results in this research open the door to new possibilities for the development of artificial intelligence in the area of analysis of blockchain transactions. The introduction of various techniques such as Gradient Boosting Classifier, Random Forest Classifier, or Autoencoder allows for flexible approaches depending on the specifics of the analyzed case. These diverse tools have the potential to adapt to the changing environment of blockchain transactions, which is crucial given the continuous development of this technology.

The presented Fuzzy Neural Network with Ordered Fuzzy Numbers has great potential for use in anomaly detection. Particularly important is the fact that this solution achieves a high operational speed [33,34].

Additionally, there are situations where the full interpretability of models is limited, which complicates the understanding of decisions made by the system. This issue faces a challenge of how to increase transparency and trust in the applied algorithms, especially in the context of a sensitive domain such as the analysis of blockchain transactions.

The need for further optimization of algorithms in terms of computational efficiency also remains significant. Contemporary blockchain environments generate vast amounts of data, and the effectiveness of the applied methods must be balanced with their speed of operation to maintain practicality and usefulness in real-world conditions.

Looking into the future, prospects for further research include expanding the analysis to various types of blockchains, not only public ones but also those implemented in a private model. The pursuit of more advanced exploration of machine learning techniques provides opportunities for more precise adaptation to the dynamic landscape of blockchain transactions.

As research progresses, there is also a need for a deeper understanding of potential challenges related to interpreting decisions made by models. In situations where full interpretability is limited, the development of strategies to increase understanding of decision-making processes becomes necessary. This is a key aspect for the acceptance and implementation of these technologies in practice, especially in critical domains such as the analysis of blockchain transactions.

Another important point to consider is the adaptability of models to different types of blockchains. This includes not only public blockchains but also those operating in closed environments. The pursuit of the universality and scalability of transaction analysis algorithms should be a priority to enable their effective application in diverse scenarios.

In terms of practical implementation, the development of monitoring systems that not only react to current anomalies but are also integrated with adaptive mechanisms are of undeniable value. Creating dynamic models that can adjust to the evolving landscape of blockchain transactions may be a crucial factor in ensuring the long-term effectiveness and usefulness of these solutions in dynamic financial and technological environments. In this field, fuzzy neural networks seem to be a promising research direction.

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